Diagonalized Low-Rank Learning for Hyperspectral Image Classification

Changda Xing, Meiling Wang, Zhisheng Wang, Chaowei Duan, and Yiliu Liu

***Abstract*— Hyperspectral image (HSI) classification is a current research hotspot. Most existing methods usually export discriminative features with low-quality distribution and low information utilization, which may induce classification performance degeneration. To remedy such deficiencies, we propose a diagonalized low-rank learning (DLRL) model for HSI classification in this study. Specifically, a classwise regularization is used to capture the classwise block-diagonal structure of low-rank representation, which can further cluster the represented HSI pixels from one class into the same subspace and extract features with well-ordered distribution. Such a regularization assists to easily and correctly classify HSIs. In addition, we combine sparsity and collaboration to extract more discriminative features for guaranteeing high information utilization, i.e., a tradeoff of sparsity and collaboration is sought to acquire both correlations among HSI pixels and characteristics of each pixel. By this way, rich information in the HSI can be fully used for good feature extraction. Further, the estimated feature representation is used as an input to the support vector machine (SVM) classifier for HSI classification. Extensive experiments have been done to validate that the proposed DLRL method achieves much classification performance in contrast to several state-of-the-art algorithms.**

***Index Terms*— Collaboration, hyperspectral image (HSI) classification, low-rank representation (LRR), sparsity.**

I. Introduction

IN HYPERSPECTRAL images (HSIs), rich information for various land covers and uses is included, which is obtained by sensors capturing the radiance of materials from hundreds of contiguous narrow spectral bands. Because of their strong information perception, HSIs have widely applied to many applications, such as forest monitoring, urban mapping, climate prediction, and ocean exploration [1]–[3]. Among these applications, the classification technique is a significant computer task, which is mainly used to classify and identify different earth surface features (such as types of crops, soil, grass, urban area *et al.*). This technique assists for human's observation and computer-based intelligent analysis.

Manuscript received December 17, 2020; revised April 22, 2021 and May 27, 2021; accepted May 28, 2021. Date of publication June 15, 2021; date of current version January 12, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 61473144, Grant 61903189, and Grant 61973158. *(Corresponding author: Meiling Wang.)*

Changda Xing, Zhisheng Wang, Chaowei Duan, and Yiliu Liu are with the College of Automation Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing 211106, China.

Meiling Wang is with the College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing 211106, China (e-mail: mely@nuaa.edu.cn).

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2021.3085672, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2021.3085672

In the past many years, numerous classification strategies for HSIs have been designed. Among these, classification strategies with spectral features are classical, such as nonparametric discriminant analysis [4], minimized noise fraction [5], locality-preserving dimensionality reduction [6], and raw spectral features only (RAW) [7]. These methods consider each endmember as a feature vector for discriminant analysis. However, they ignore the spatial characteristics, which leads to classification performance degeneration. To remedy such deficiency, many classification strategies with spatial features have been designed, which further enhance the separability of HSI information. For example, literature [8] has presented an algorithm based on gray level co-occurrence matrix (GLCM) to extract spatial features. It can effectively describe the texture information of HSIs. Literature [9] presents the famous feature extraction method with the multiscale morphology. This technique combines geodesic opening and closing operations of different sizes to build a differential morphological profile that records image structural information. Motivated by such idea, many improved morphology-based methods have been designed, such as extended morphological attribute profiles (EMAP) [10], directional morphological profiles [11], and morphological profiles with differently shaped structuring elements [12]. Other representative methods are based on wavelet transform [13], empirical mode decomposition [14], and tensor [15]

In the last decade, deep learning has attracted extensive attention in the field of HSI classification. It benefits from the strong ability to well describe abstract and complex structure information. Therefore, deep learning-based classification methods have great robustness and invariance toward local changes of HSIs, which have achieved competitive performance [16]–[26]. For example, in [16], the concept of deep learning is for the first time introduced into HSI classification. Stacked autoencoders (SAEs) are adopted to get useful high-level features for the logistic regression classifier. Literature [17] first uses the principal component analysis (PCA) to reduce the dimensionality of the input HSI, and then computes the spatial information in neighborhood regions by a 2-D convolutional neural network (CNN). This method can not only extract the discriminative spatial features but reduce the computational cost. More advanced, Xu *et al.* [18] have utilized a random patches network (RPNet) to extract both shallow and deep convolutional features, and these two types of features are combined for HSI classification. Cao *et al.* [19] have presented a CNN-active learning (AL)-Markov random field (MRF), which integrates both AL and deep network

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:04:11 UTC from IEEE Xplore. Restrictions apply.

into an unified framework to improve the classification performance.

More recently, representation learning has announced impressive results and opened up new avenues for the HSI classification. In general, representation learning strategies for classification mainly include sparse representation (SR) and low-rank representation (LRR). For SR-based methods, an unknown testing sample can be approximately represented by the combination of a few training samples from the entire dictionary, whereas the corresponding sparse vector encodes the class information implicitly. Some popular SR-based methods have the joint sparsity model [27], group sparse coding [28], Laplacian regularized Lasso [29], and multiple-feature-based adaptive SR [30]. For LRR-based methods, they are dedicated to seek a data representation under lowest-rank constraint and provide a reliable way for capturing the global structure of HSIs. For example, in [31], He *et al.* have proposed a low-rank group-inspired dictionary learning (LGIDL) for HSI classification, in which structured dictionary is learned to capture global structure of the given HSI. In [32], group-based sparse and LRRs are learned, which facilitate to identify both local and global structure of the HSI. In [33], Chen *et al.* have proposed a low-rank discriminative least squares regression model (LRDLSR) for HSI classification. This method class wisely imposes low-rank constraint on the relaxed labels obtained by nonnegative relaxation matrix to improve its within-class compactness and similarity.

It should be noted that two motivations are included in this study. *First*, the ideal feature representations for HSI classification is the best to have classwise block-diagonal structure that is of well-ordered distribution, which means spectral pixels of the same class are represented by the base elements of corresponding class. However, most previous works do not include such structure and exploit features with disordered distribution or low distribution quality for HSI classification. In such feature space, a feature vector may be very close to ones from other classes [34], which will increase the probability of classification errors. Therefore, how to obtain discriminative feature representation with the classwise block-diagonal structure is the first motivation of our study. *Second*, as shown in [27]–[30], the SR-based methods have the advantage of representing each pixels from the HSI individually. Whereas these methods ignore correlations among pixels. Fortunately, such correlations can be described using the l_2 norm collaboration, i.e., collaborative representation (CR). Note that the CR model has less sparsity and low ability to represent each HSI pixels. Hence, seeking a tradeoff of sparsity with l_1 norm and collaboration with l_2 norm to extract more discriminative features for high-quality HSI classification is the second motivation of our study.

Considering the motivations listed above, we propose a novel method based on diagonalized low-rank learning (DLRL) for HSI classification in this study. Specifically, a classwise regularization is exploited to capture the classwise block-diagonal structure of LRR, which can further cluster the represented HSI pixels from one class into the same subspace and extract features with well-ordered distribution. Such way assists to easily and correctly classify HSIs. In addition, considering correlations among HSI pixels and sparsity prior, the Frobenius norm constraint and l_1 norm constraint are introduced to seek a tradeoff of collaboration and sparsity. Thus, we can fully use HSI information for good feature extraction. Further, the estimated feature representation is used as an input to the support vector machine (SVM) classifier for HSI classification. We evaluate the proposed DLRL method on three hyperspectral datasets including Salines, Pavia Center, and Botswana, with experimental results validating its effectiveness in HSI classification.

The major contributions and novelties of this article are threefold: 1) We first propose a method with DLRL, which considers both feature distribution and information utilization to extract high-quality feature representation. The estimated feature representation is then used as an input to the SVM classifier for HSI classification. 2) A classwise regularization is introduced to capture the classwise block-diagonal structure of LRR. By such way, the represented HSI features have well-ordered distribution used for correct HSI classification. 3) A tradeoff of collaboration and sparsity is sought to obtain both correlations among HSI pixels and characteristics of each pixel, which aims to fully use HSI information for high-quality classification.

The remainder of this article is organized as follows: In Section II, we briefly show related works and motivations of our study. The proposed method for HSI classification is introduced in Section III. In Section IV, experimental results on three datasets and related discussion are presented. We finally conclude this article in Section V.

II. Related Works and Motivations

In this section, we first provide an overview of LRR, SR, and CR, which are closely related to our study. Second, motivations of our study are presented.

A. Low-Rank Representation

LRR aims to find the lowest-rank representation of data. This technique can effectively capture the global structure of data, and such structural information plays an important role in data analysis. LRR was initially proposed by Liu *et al.* in [35] to solve subspace clustering problems. After that, LRR has achieved successes in many applications, such as objective detection and image classification. For example, in [36], Qu *et al.* proposed a generalized LRR form to solve estimation problem. In [37], the maximum a posterior model is constructed upon LRR for HSI segmentation. Zhao and Yang [38] added LRR regularization into the HSI denoising model to deal with the global redundancy and correlation in spectral domain. As we know, high spatial resolution is included in HSIs, and there exists high similarity among different spectral pixel vectors from the same neighborhood. This implies the low-rank characteristic of data. Therefore, LRR provides a feasible tool to carry out HSI analysis. Lots of LRR-based methods have been developed in recent years. For example, He *et al.* [32] designed LRR-based model to identify both local and global structure of data for HSI classification. In [33], Chen *et al.* proposed a LRDLSR for HSI classification.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:04:11 UTC from IEEE Xplore. Restrictions apply.

B. SR With l_1 Norm

As a widely utilized tool, SR devotes to find a suitable way that focused data can be represented or approximately represented as a linear combination of a "few" atoms from redundant dictionary [39]. Such representation form is achieved by imposing l_1 norm on objective variables. For HSIs, numerous redundancy is contained, which means the sparsity of data. Therefore, the SR provides a reliable technique to solve the problem of HSI classification. Many SR-based methods have been demonstrated for superiority in literature [27]–[30]. For instance, Zhang *et al.* [27] designed the joint sparsity model for HSI classification. It assumes that the sparse vectors of all the neighboring pixels share a common sparsity support. Hence, the corresponding representation matrix has a row sparsity pattern. In [28], Zhang *et al.* introduced group sparse coding to process each spectral pixel and its spatial neighbors together for obtaining a smooth representation, which can improve the separability of HSIs. Chen *et al.* [29] used Laplacian regularized Lasso to characterize the similarity among neighboring pixels by an introduced weight matrix. In [30], Fang *et al.* proposed an adaptive SR for HSI classification, in which four different features are extracted to be sparsely coded for determining the class label of each test pixel.

C. CR With Frobenius Norm

Different from SR with l_1 norm regularization, which finds sparsest representation of each samples individually, the CR can investigate much the role of collaboration reflecting correlations among the query samples. Such argument has been validated in literature [2], [40]. The collaboration is achieved by imposing l_2 norm on objective variables, and which brings some discriminative information for vision tasks. Because the l_2 norm has consistency with Frobenius norm, the above-mentioned collaboration can be also determined by Frobenius norm. Although the CR has good ability to describe correlations by using l_2 norm, sparsity is much weaker than that by SR with l_1 norm. Because of rich correlations contained in HSIs, CR has great potential to extract more discriminative information for HSI classification.

D. Motivations

For the task of HSI classification, if the discriminative features for classifiers are of well-ordered distribution, the classification accuracy will be improved. The ideal feature representations of all spectral pixels is the best to have classwise block-diagonal structure, which means spectral pixels of the same class are represented by the base elements of a corresponding class. It should be noted that most of the previous works for HSI classification do not include such block-diagonal structure, which export discriminative features with disordered distribution or low distribution quality. In such feature space, a feature vector may be very close to the ones from other classes [34], which will increase the probability of classification errors. Therefore, we attempt to construct a model that can obtain discriminative feature representation with the classwise block-diagonal structure, which is the first motivation of our study.

In addition, the SR-based methods have advantage of representing each pixel from the HSI individually. In fact, except for spectral information within pixels, HSI data also contain a lot of correlation information among different spectral pixels, which is difficult to be described by SR. Fortunately, such correlations can be described using the l_2 norm collaboration, i.e., CR. However, note that the weaker sparsity is included in the CR model, and it is hard to guarantee good accuracy. Therefore, we will seek a tradeoff of sparsity with l_1 norm and collaboration with l_2 norm to guarantee high information utilization and extract more discriminative features for high-quality HSI classification, which is the second motivation of our study.

III. Methodology

In this study, we develop a DLRL for the classification of HSIs. The details of this method are shown in this section. Here, we have some illustrations as follows: $\mathbf{X} \in \mathbb{R}^{L\times M\times N}$ denotes the given HSI. For convenient discussion, it is unfolded into a 2-D matrix $X = \{x_1, x_2, \ldots, x_k, \ldots, x_K\} \in \mathbb{R}^{L\times K}$, where $K = M \times N$ and x_k denotes pixel vector of the given HSI $\mathbf{X}$. We define $X = [A, B]$, where $A \in \mathbb{R}^{L\times K_a}$ and $B \in \mathbb{R}^{L\times K_b}$ are training and testing samples, respectively.

A. Problem Statement and Model Construction

In the problem of HSI classification, the low-rank learning-based methods have been proposed. These approaches are dedicated to find the lowest-rank representation of the given X by a dictionary, whose mathematical model can be described as follows:

$$\arg\min_{Z} \ \|Z\|_* \quad \text{s.t. } X = DZ \tag{1}$$

where $\|Z\|_*$ is the nuclear norm of Z, and D denotes a dictionary. It should be noted that previous methods with model (1) and its extends exported features with unideal distribution. In such feature space, a feature vector may be very close to the ones from other classes, which will reduce the accuracy of classifiers. Therefore, we attempt to find a low-rank feature representation form with the classwise block-diagonal structure, in which the distribution of the represented features is well ordered, and the corresponding spectral pixels are represented by the base elements of the same class. More specifically, for a training sample set A, we expect its representation form ${}_aZ$ as follows:

$${}_aZ = \begin{bmatrix} {}_aZ_1^* & 0 & 0 & 0 \\ 0 & {}_aZ_2^* & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & {}_aZ_C^* \end{bmatrix}. \tag{2}$$

To capture the above-described classwise block-diagonal structure, a classwise regularization $\sum_{i=1}^{C} \|{}_aZ_{\bullet i}\|_F^2$ is introduced, where $\|\cdot\|_*$ is the Frobenius norm operator; ${}_aZ_{\bullet i}$ is the representation matrix of training samples from the ith class subject to $A_{\bullet i}$. The training sample matrix A is here viewed as the dictionary, and $A_{\bullet i}$ denotes the matrix excluding the ith class training samples. By exploiting such regularization,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:04:11 UTC from IEEE Xplore. Restrictions apply.

the training samples can be represented by the bases of corresponding class. Furthermore, Zhang *et al.* [40] insisted that the collaboration with Frobenius or l_2 norm constraints can investigate much the correlations among samples. To fully use HSI information for classification, a tradeoff of collaboration and sparsity can be found to acquire both correlations among HSI pixels and characteristics of each pixel. Suggested by the above-presented arguments, a DLRL is built into

$$\begin{aligned}\arg\min_{Z,E}\ & \|Z\|_* + \frac{\alpha}{2}\sum_{i=1}^{C}\|{}_aZ_{\bullet i}\|_F^2 + \frac{\beta}{2}\|Z\|_F^2 + \gamma\|Z\|_1 \\ & + \lambda\|E\|_1 \\ \text{s.t. } & X = AZ + E\end{aligned} \tag{3}$$

where α, β, γ, and λ are regularization parameters; E denotes the sparse noise matrix; $\|\cdot\|_1$ denotes the l_1-norm operator; and ${}_bZ$ is the representation matrix of testing samples. For model (3), the classwise regularization guarantees to obtain the low-rank feature representation with the classwise block-diagonal structure. The Frobenius norm regularization $||Z||_F^2$ is used to extract the correlations among HSI pixels. The l_1-norm regularization $||Z||_1$ has a good ability to describe the characteristics of each pixel and suppress noises.

B. Optimization Strategy

As can be found in (3), two variables Z and E are included, and such objective function is convex for each variable once fixing the other variable. Therefore, we employ the proximal alternating optimization scheme to solve (3), which can be guaranteed to converge to a critical point under certain conditions. Two auxiliary variables $\mathcal{P}$ and $\mathcal{Q}$ are here introduced, and problem (3) is equivalent to the following form:

$$\begin{aligned}\arg\min_{Z,E,\mathcal{P},\mathcal{Q}}\ & \|\mathcal{P}\|_* + \frac{\alpha}{2}\sum_{i=1}^{C}\|{}_aZ_{\bullet i}\|_F^2 + \frac{\beta}{2}\|Z\|_F^2 + \gamma\|\mathcal{Q}\|_1 \\ & + \lambda\|E\|_1 \\ \text{s.t. } & X = AZ + E,\quad Z = \mathcal{P},\ Z = \mathcal{Q}.\end{aligned} \tag{4}$$

The problem form (4) can be solved by the Augmented Lagrange Multiplier (ALM) method, and its augmented Lagrange function is written into

$$\begin{aligned}&\mathcal{L}(Z, E, \mathcal{P}, \mathcal{Q}, \mathcal{Y}_1, \mathcal{Y}_2, \mathcal{Y}_3) \\ &= \|\mathcal{P}\|_* + \frac{\alpha}{2}\sum_{i=1}^{C}\|{}_aZ_{\bullet i}\|_F^2 + \frac{\beta}{2}\|Z\|_F^2 + \gamma\|\mathcal{Q}\|_1 \\ &\quad + \lambda\|E\|_1 + \mathbf{tr}\big(\mathcal{Y}_1^T(X - AZ - E)\big) + \mathbf{tr}\big(\mathcal{Y}_2^T(Z - \mathcal{P})\big) \\ &\quad + \mathbf{tr}\big(\mathcal{Y}_3^T(Z - \mathcal{Q})\big) \\ &\quad + \frac{\mu}{2}\big(\|X - AZ - E\|_F^2 + \|Z - \mathcal{P}\|_F^2 + \|Z - \mathcal{Q}\|_F^2\big)\end{aligned} \tag{5}$$

where $\mu > 0$ represents the penalty parameter; $\mathcal{Y}_1$, $\mathcal{Y}_2$, and $\mathcal{Y}_3$ are the matrices of Lagrange multipliers; $\mathbf{tr}(\cdot)$ denotes the trace of a matrix. We have the optimization process as follows.

1) Optimization With Respect to $\mathcal{P}$: In the augmented Lagrange function (5), fixing variables Z, E, $\mathcal{Q}$, $\mathcal{Y}_1$, $\mathcal{Y}_2$, and $\mathcal{Y}_3$, we can update variable $\mathcal{P}$ by the following problem:

$$\begin{aligned}&\mathcal{P}^{j+1} \\ &= \arg\min_{\mathcal{P}}\ \|\mathcal{P}\|_* + \mathbf{tr}\big(\big[\mathcal{Y}_2^j\big]^T(Z^j - \mathcal{P})\big) + \frac{\mu}{2}\|Z^j - \mathcal{P}\|_F^2 \\ &= \arg\min_{\mathcal{P}}\ \frac{1}{\mu}\|\mathcal{P}\|_* + \frac{1}{2}\big\|\mathcal{P} - \big(Z^k + \mathcal{Y}_2^j/\mu\big)\big\|_F^2 \\ &= \mathbf{max}\big(\mathcal{S}_{1/\mu}\big(Z^k + \mathcal{Y}_2^j/\mu\big), 0\big)\end{aligned} \tag{6}$$

where $[\mathcal{Y}_2^j]$ and Z^j are the updated values of $\mathcal{Y}_2$ and Z at the jth iteration; $\mathbf{max}(x, 0)$ is used to select the maximum one between x and 0; $\mathcal{S}_{1/\mu}(\cdot)$ is the singular value thresholding operator [36], [41]. Specifically, assume that $(Z^k + \mathcal{Y}_2^j/\mu)$ has rank t, and which includes the singular value decomposition form $(Z^k + \mathcal{Y}_2^j/\mu) = U\Lambda V^T$, where $\Lambda = \mathbf{diag}(\{\pi_i\}_{1\le i\le t})$ and $\mathbf{diag}(\cdot)$ denotes a diagonal matrix. Then, the form of $\mathcal{S}_{1/\mu}$ can be written as

$$\begin{aligned}\mathcal{S}_{1/\mu}\big(Z^k + \mathcal{Y}_2^j/\mu\big) &= U\mathcal{S}_{1/\mu}(\Lambda)V^T \\ \mathcal{S}_{1/\mu}(\Lambda) &= \mathbf{diag}\left(\mathbf{max}\left(\left\{\pi_i - \frac{1}{\mu}\right\}_{1\le i\le t}\right), 0\right).\end{aligned} \tag{7}$$

2) Optimization With Respect to $\mathcal{Q}$: Keeping with Z, E, $\mathcal{P}$, $\mathcal{Y}_1$, $\mathcal{Y}_2$, and $\mathcal{Y}_3$ fixed in (5), the updated values of $\mathcal{Q}$ can be calculated by

$$\begin{aligned}&\mathcal{Q}^{j+1} \\ &= \arg\min_{\mathcal{Q}}\ \gamma\|\mathcal{Q}\|_1 + \mathbf{tr}\big(\big[\mathcal{Y}_3^j\big]^T(Z^j - \mathcal{Q})\big) + \frac{\mu}{2}\|Z^j - \mathcal{Q}\|_F^2 \\ &= \arg\min_{\mathcal{Q}}\ \frac{\gamma}{\mu}\|\mathcal{Q}\|_1 + \frac{1}{2}\big\|\mathcal{Q} - \big(Z^k + \mathcal{Y}_3^j/\mu\big)\big\|_F^2 \\ &= S_{\gamma/\mu}\big(Z^k + \mathcal{Y}_3^j/\mu\big)\end{aligned} \tag{8}$$

where $[\mathcal{Y}_3^j]$ is the updated value of $\mathcal{Y}_3$ at the jth iteration; $S_{\gamma/\mu}(\cdot)$ is the shrinkage operator, whose form is given as following:

$$S_{\gamma/\mu}(\cdot) = \mathbf{sign}(\cdot)\mathbf{max}\left(|\cdot| - \frac{\gamma}{\mu}, 0\right) \tag{9}$$

where $\mathbf{sign}(x)$ is the sign function.

3) Optimization With Respect to Z: Fixing variables E, $\mathcal{P}$, $\mathcal{Q}$, $\mathcal{Y}_1$, and $\mathcal{Y}_2$ in (5), the Z can be computed by solving the optimization problem as follows:

$$\begin{aligned}&Z^{j+1} \\ &= \arg\min_{Z}\ \frac{\alpha}{2}\sum_{i=1}^{C}\|{}_aZ_{\bullet i}\|_F^2 + \frac{\beta}{2}\|Z\|_F^2 \\ &\quad + \mathbf{tr}\big(\big[\mathcal{Y}_1^j\big]^T(X - AZ - E^j)\big) + \mathbf{tr}\big(\big[\mathcal{Y}_2^j\big]^T(Z - \mathcal{P}^{j+1})\big) \\ &\quad + \mathbf{tr}\big(\big[\mathcal{Y}_3^j\big]^T(Z - \mathcal{Q}^{j+1})\big) + \frac{\mu}{2}\|X - AZ - E^j\|_F^2 \\ &\quad + \frac{\mu}{2}\|Z - \mathcal{P}^{j+1}\|_F^2 + \frac{\mu}{2}\|Z - \mathcal{Q}^{j+1}\|_F^2\end{aligned} \tag{10}$$

where E^j is the updated value of E at the jth iteration. For solving problem (10), we here construct a matrix $\mathcal{C}$, which has

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:04:11 UTC from IEEE Xplore. Restrictions apply.

the following form:

$$\mathcal{C} = [{}_a\mathcal{C}^j, {}_b\mathcal{C}^j] = \left[\begin{bmatrix} {}_aZ_1^j & 0 & 0 & 0 \\ 0 & {}_aZ_2^j & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & {}_aZ_C^j \end{bmatrix}, {}_b\mathcal{C}^j\right] \tag{11}$$

where ${}_a\mathcal{C}^j$ and ${}_b\mathcal{C}^j$ are submatrix of Z^j, which, respectively, corresponds to the representations of training set A and testing set B at the jth iteration. Problem (11) is thus converted into

$$\begin{aligned} Z^{j+1} = \arg\min_{Z}\ & \frac{\alpha}{2}\|Z - {}_b\mathcal{C}^j\|_F^2 + + \frac{\beta}{2}\|Z\|_F^2 \\ & + \mathbf{tr}\big(\big[\mathcal{Y}_1^j\big]^T(X - AZ - E^j)\big) + \mathbf{tr}\big(\big[\mathcal{Y}_2^j\big]^T(Z - \mathcal{P}^{j+1})\big) \\ & + \mathbf{tr}\big(\big[\mathcal{Y}_3^j\big]^T(Z - \mathcal{Q}^{j+1})\big) + \frac{\mu}{2}\|X - AZ - E^j\|_F^2 \\ & + \frac{\mu}{2}\|Z - \mathcal{P}^{j+1}\|_F^2 + \frac{\mu}{2}\|Z - \mathcal{Q}^{j+1}\|_F^2. \end{aligned} \tag{12}$$

Problem (12) is convex, therefore the optimal solution is directly given as follows:

$$\begin{aligned} Z^{j+1} = & ((\alpha/\mu + \beta/\mu + 2)\mathbf{I} + A^TA)^{-1} \\ & \times \big(A^T(X - E^j) + \mathcal{P}^{j+1} + \mathcal{Q}^{j+1} \\ & + \big(\alpha_b\mathcal{C}^j + A^T\mathcal{Y}_1^j - \mathcal{Y}_2^j - \mathcal{Y}_3^j\big)/\mu\big). \end{aligned} \tag{13}$$

The (13) has the complexity of $O(L^3)$ and suffers from the heavy computation. We introduce the Sherman-Morrison method [42] to reduce such complexity to $O(L^2k_a)$, which can guarantee the fast computation of (13).

4) Optimization With Respect to E: We fix variables Z, $\mathcal{P}$, $\mathcal{Q}$, $\mathcal{Y}_1$, and $\mathcal{Y}_2$ in (5), and update variable E by the problem form as follows:

$$\begin{aligned} E^{j+1} = & \arg\min_{E}\ \lambda\|E\|_1 + \mathbf{tr}\big(\big[\mathcal{Y}_1^j\big]^T(X - AZ^{j+1} - E)\big) \\ & + \frac{\mu}{2}\|X - AZ^{j+1} - E\|_F^2 \\ = & \arg\min_{E}\ \frac{\lambda}{\mu}\|E\|_1 + \frac{1}{2}\big\|E - \big(X - AZ^{j+1} + \mathcal{Y}_1^j/\mu\big)\big\|_F^2 \\ = & S_{\lambda/\mu}\big(X - AZ^{j+1} + \mathcal{Y}_1^j/\mu\big) \end{aligned} \tag{14}$$

where $S_{\lambda/\mu}(\cdot)$ is the shrinkage operator, whose mathematical form is seen in (9).

5) Updating Multipliers $\mathcal{Y}_1^j$, $\mathcal{Y}_1^j$, and $\mathcal{Y}_1^j$ in (5): The augmented Lagrange function (5) includes three multipliers $\mathcal{Y}_1^j$, $\mathcal{Y}_1^j$, and $\mathcal{Y}_1^j$. To update these variables, we have the following forms:

$$\mathcal{Y}_1^{j+1} = \mathcal{Y}_1^j + \mu(X - AZ^{j+1} - E^{j+1}) \tag{15}$$

$$\mathcal{Y}_2^{j+1} = \mathcal{Y}_2^j + \mu(Z^{j+1} - \mathcal{P}^{j+1}) \tag{16}$$

$$\mathcal{Y}_3^{j+1} = \mathcal{Y}_3^j + \mu(Z^{j+1} - \mathcal{Q}^{j+1}). \tag{17}$$

These iterative updates will be stopped once satisfying criterion conditions for convergence. Such criteria can be represented by

$$\frac{\|X - AZ - E\|_F^2}{\|X\|_F^2} < \epsilon_1, \quad \frac{\|Z - P\|_F^2}{\|Z\|_F^2} < \epsilon_2, \quad \frac{\|Z - Q\|_F^2}{\|Z\|_F^2} < \epsilon_3 \tag{18}$$

where ϵ_1, ϵ_2, and ϵ_3 are small constants, which are all set to 10^{-7} in our study.

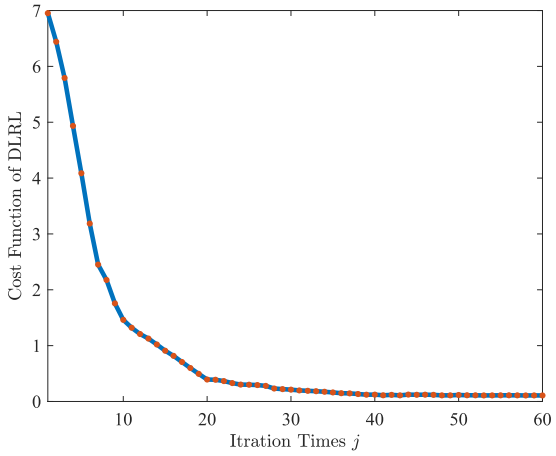


Fig. 1. Evolution of the cost function of DLRL on iteration times j.

By the above-presented optimization process, the optimal LRR Z is obtained. Here, an example is used to visually present the evolution of the cost function of DLRL to observe our optimization strategy in Fig. 1. As can be found in this figure, the value of the cost function of our DLRL is monotonically decreasing with the increasing of iteration times, and the optimization convergence is almost achieved after about 50 iterations. In other words, the cost function is minimized after 50 iterations. Such results can verify that our proposed optimization strategy is effective and efficient.

C. SVM Classifiers for Low-Rank Feature Representation Z

We here recall HSI data $X = [A, B]$, where $A \in \mathbb{R}^{L\times K_a}$ and $B \in \mathbb{R}^{L\times K_b}$ are training and testing samples. Using our designed model in (3), the low-rank feature representation $Z = [{}_aZ, {}_bZ]$ can be extracted, where ${}_aZ$ and ${}_bZ$ are denoted by LRRs corresponding A and B, respectively. We exploit multiple binary SVM classifiers to classify such feature representation Z with C classes, which is achieved by building one-versus-rest (OVR) SVMs [43]. Specifically, to build the OVR SVMs, we consider when training the samples from one class are viewed into a class, and remaining samples from other classes are grouped into another class. For convenient understanding, suppose that C classes from ${}_aZ$ are denoted by $[L_1, L_2, \ldots, L_C]$, and C train sets $[S_1, S_2, \ldots, S_C]$ can be determined by the following way:

S_1: L_1 is viewed as a "1" set; $L_2 \sim L_C$ are viewed as a "0" set.

S_2: L_2 is viewed as a "1" set; $L_1, L_3 \sim L_C$ are viewed as a "0" set.

$\vdots$

S_C: L_C is viewed as a "1" set; $L_1 \sim L_{C-1}$ are viewed as a "0" set.

With these C train sets, C binary SVMs are trained, where each binary SVM is trained by the following form:

$$\begin{aligned} & \arg\max_{d_c}\ \sum_{i=1}^{K_a} d_{c,i} - \frac{1}{2}\sum_{i=1}^{K_a}\sum_{j=1}^{K_a} d_{c,i}d_{c,j}\mathcal{A}_{c,i}\mathcal{A}_{c,j}\mathcal{K}(S_{c,i}, S_{c,j}) \\ & \text{s.t.}\ \sum_{i=1}^{K_a} D_{c,i}\mathcal{A}_{c,i} = 0, \quad d_{c,i} \geq 0,\ 1 \leq c \leq C \end{aligned} \tag{19}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:04:11 UTC from IEEE Xplore. Restrictions apply.

where d_c is the Lagrange coefficient for set S_c; K_a represents the number of training samples; $\mathcal{K}(\cdot,\cdot)$ is the kernel function of SVM.

Once C SVMs are successfully trained, we can use them to predict any unknown samples, and the corresponding classification result is determined by selecting label "1" among C SVM values. Each SVM value is computed by

$$\text{label}(z) = \textbf{sign}\left(\sum_{i=1}^{K_a} d_{c,i}\mathcal{A}_{c,i}\mathcal{K}(S_{c,i}, z) + h_c\right) \tag{20}$$

where z is a LRR feature from ${}_bZ$ and label(z) denotes the predicted label of z; h_c is the decision offset coefficient that can be determined by the trained d_c, following the Karush–Kuhn–Tucker (KKT) condition [44]. By the above SVM classifiers, we can classify samples to obtain class labels.

Three points are here remarked for the proposed DLRL classification method. *First*, the classwise block-diagonal structure is designed by introducing a classwise regularization $\sum_{i=1}^{C} \|_aZ_{\bullet i}\|_F^2$. By such way, the LRR features are diagonalized to be more discriminative. *Second*, Frobenius norm and l_1 norm constraints are introduced to seek a tradeoff of collaboration and sparsity, which aims to fully use both correlations among HSI pixels and characteristics of each pixel for high-quality classification. *Third*, an optimization strategy based on proximal alternating optimization scheme is constructed to obtain solution Z of LRR, and an example is adopted to validate whether such scheme is effective and efficient.

IV. Experiment

A. Experimental Datasets

For evaluating the performance of the proposed DLRL classification method, we perform experiments on three datasets including Salinas, Pavia Center, and Botswana, which have also been widely used in many related works [16]–[31]. These data can be downloaded at website http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes. The Salinas dataset is collected with a 512 × 217 size by the 224-band Airborne/Visible Infrared Imaging Spectrometer (AVIRIS) sensor over Salinas Valley at California. The 20 water-absorption bands are also discarded for reducing the number of bands to 204. To decrease computational cost, we down-sample such data by a 2 factor to obtain bands with a 256 × 108 size. There are 16 classes in the corresponding ground truth. The second dataset named as Pavia Center is recorded by the reflective optics system-imaging spectrometer (ROSIS) sensor over the Center of Pavia area. The spectrum ranges are covered from 430 nm to 860 nm with an interval of 4 nm. We remain 103 bands by abandoning noise bands, and which is down-sampled by a 3 factor to obtain the spatial resolution of 365 × 238; 9 classes are included in the ground truth. The third data are called as Botswana, which was a sequence of data acquired using the NASA EO-1 satellite over the Okavango Delta, Botswana, in 2001. Its size lies in 1476 × 256 × 145, and it consists of observations from 14 identified classes representing the land cover types in seasonal swamps, occasional swamps, and drier woodlands located in the distal portion of the Delta.

B. Quantitative Metrics

In our experiments, three widely used evaluation metrics to measure the quality of classification results of different methods were as follows: overall accuracy (OA), average accuracy (AA), and Kappa coefficient. The OA is the percentage of pixels that are properly categorized. AA is the average percentage of pixels for each class. The Kappa coefficient corrects the correct percentage of classified pixels. In this experimental section, all listed metric results were obtained by averaging over five experiments.

C. Parameter Analysis

To well investigate the performance of the developed DLRL classification method, it is essential to analyze the selection of some critical parameters. In model (3), parameter α is a weight used to control the contribution of the classwise regularization $\sum_{i=1}^{C} \|_aZ_{\bullet i}\|_F^2$. The β denotes a regularization parameter to tune the collaboration with Frobenuis norm in (3). The γ and λ are adopted to regulate the sparsity of LRR and noises. The μ is a penalty parameter in (5), which is used to couple the objective function and three equation constraints in (4). We will perform experiments to explore the effects of these five parameters on the DLRL classification performance. The Salinas dataset is here selected as a testing example. Metrics OA and Kappa are here used as the evaluation indexes. The influences of parameters α, β, γ, and λ on DLRL for OA and Kappa are visually plotted in Fig. 2.

1) Analysis of Classwise Regularization Parameter α: Fig. 2(a) is adopted to show overall classification performance (in OA and Kappa) affected by classwise regularization parameter α. As can be seen, OA and Kappa achieve the best values at $\alpha = 0.6$. When α is lower than 0.6, the values of OA and Kappa are relatively small. The reason is that a small α value leads to reduction in the contribution of the classwise regularization $\sum_{i=1}^{C} \|_aZ_{\bullet i}\|_F^2$. When the α value is increased over 0.6 (**i**.e., $\alpha > 0.6$), which causes the degradation of LRR, the values of OA and Kappa are also prone to be decreased.

2) Analysis of Collaboration Regularization Parameter β: As illustrated in Fig. 2(b), collaboration regularization parameter β has obvious influence on the classification performance of the proposed DLRL method (in OA and Kappa). As can be found that DLRL achieves the best metric result at $\beta = 0.04$. Whereas when $\beta < 0.04$ and $\beta > 0.04$, the OA and Kappa values tend to be small, which means the worst performance of DLRL. The reason for this is that, a much small β value indicates a weak collaboration along with bad extraction of correlations among different HSI pixels. Too strong collaboration caused by over-large β values will damage the low-rank quality.

3) Analysis of Sparse Regularization Parameter γ and λ: Parameters γ and λ are employed to control the sparsity of LRR Z and noises E, whose influence on the DLRL classification performance is shown in Fig. 2(c) and (d). From these two subfigures, we can observe that γ and λ have sharp

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:04:11 UTC from IEEE Xplore. Restrictions apply.

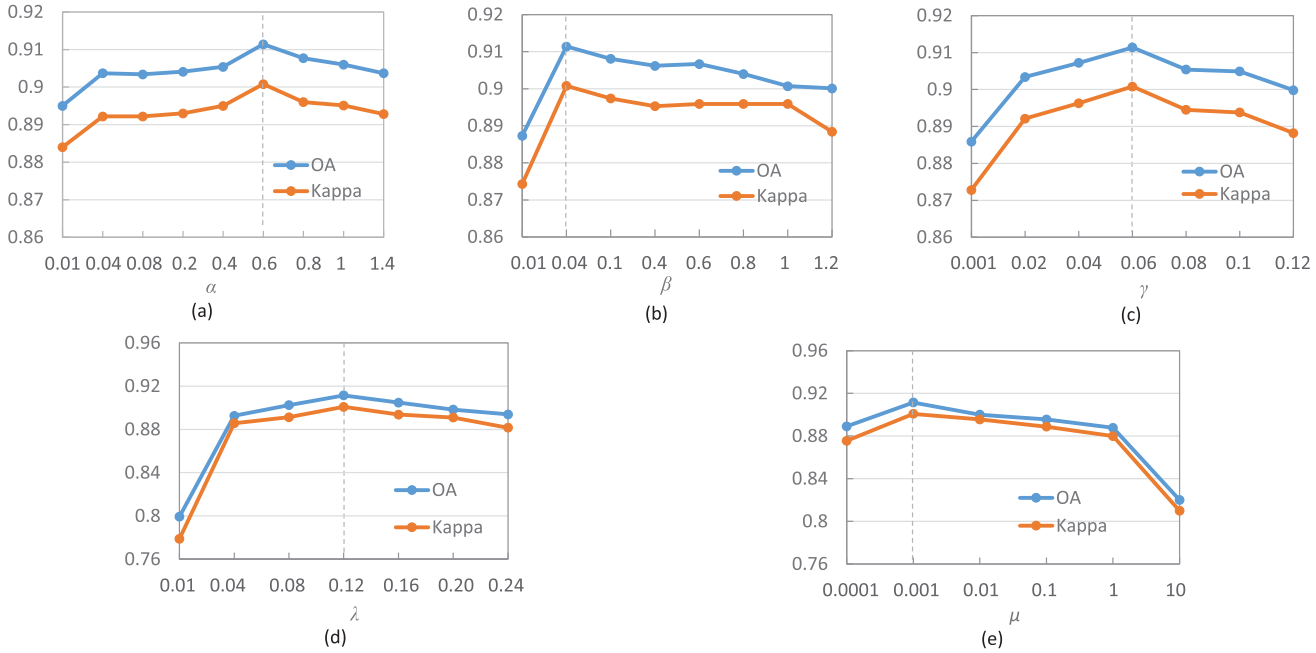


Fig. 2. Parameter influence of α, β, γ, λ, and μ on DLRL for metrics OA and Kappa. OA and Kappa curves by tuning (a) α, (b) β, (c) γ, (d) λ, and (e) μ.

influence on DLRL, and the OA and Kappa indexes achieve the best results at $\gamma = 0.06$ and $\lambda = 0.12$. Small γ and λ values (such as $\gamma = 0.001$ and $\lambda = 0.01$) weaken the sparsity of LRR and noises, which yields worse classification accuracy. Furthermore, the values of OA and Kappa are also prone to become decreasing once γ and λ lie in over-large values (such as $\gamma = 1.2$ and $\lambda = 0.28$), because excessive weight rising for the sparsity parameters γ and λ makes LRR with low quality.

4) Analysis of Penalty Parameter μ: As shown in Fig. 2(e), the μ has a significant impact on the performance of the proposed DLRL method. More specifically, we can find that the curves of OA and Kappa are convex in the interval of $[0.0001, 10]$, and which achieve the largest OA and Kappa results at $\mu = 0.001$. Once $\mu < 0.001$ and $\mu > 0.001$, the OA and Kappa values will be decreased. The reason is that, if μ is too small, such as $\mu < 0.0001$, the importance of three equation constraints will be weakened in (5). If over-large μ values are used, the term weights from the objective function in (4) are relatively lessened, which will degenerate the performance of the proposed DLRL approach.

Illuminated by the above-mentioned experimental results and related analysis, we argue that a suitable selection of parameters α, β, γ, λ, and μ would prompt the classification performance of DLRL. $\alpha = 0.6$, $\beta = 0.04$, $\gamma = 0.06$, $\lambda = 0.12$, and $\mu = 0.001$ are here selected as the best value choices in the Salinas dataset.

D. Algorithm Comparison

In experiments, six state-of-the-art classification strategies are compared to validate the superiority of the proposed DLRL method, including RAW [7], EMAP [45], image fusion and recursive filtering (IFRF) [46], LGIDL [31], LRDLSR [33], RPNet [18], and CNN-AL-MRF [19]. Among these compared algorithms, the RAW method only uses original spectral features of HSIs, and which includes the SVM classifier to achieve the HSI classification. The EMAP constructs the morphological profiles with the first three principal components for feature extraction, and the SVM classifier is also used in the EMAP. The compared IFRF method uses IFRF to extract HSI features, which are classified by SVM. The LGIDL and LRDLSR are low-rank learning based, which export low-rank feature representation also classified by SVM. The RPNet classification strategy builds RPNet with CNN to extract deep features, in which the SVM is here used as the classifier. The CNN-AL-MRF adopts the CNN and AL to construct the feature map, and the MRF strategy is further used to determine final class labels. We here give some critical parameter settings as follows: suggested by previous parameter analysis, α, β, γ, λ, and μ are set to 0.6, 0.04, 0.06, 0.12, and 0.001, respectively, in the proposed DLRL method. In addition, the default parameters of RAW, EMAP, LGIDL, LRDLSR, RPNet, and CNN-AL-MRF are given in literature [7], [45], [46], [31], [33], [18], and [19], respectively. Experimental results and related discussion are presented as follows.

We present the classification maps of RAW, EMAP, IFRF, LGIDL, LRDLSR, RPNet, CNN-AL-MRF, and proposed DLRL on the Salinas in Fig. 3, as well as Pavia Center and Botswana in Figs. S1 and S2 of ***Supplementary Materials***. As can be found in these three figures, the classification result maps of our DLRL method are more closely related to ground truth compared with those of RAW, EMAP, IFRF, LGIDL, LRDLSR, RPNet, and CNN-AL-MRF. For example, on the "Grapes untrained" and "Vineyard untrained" classes of classification maps in Fig. 3, the total correct number in Fig. 3(j) obtained by DLRL is more than those in Fig. 3(c)–(i). To quantitatively evaluate this situation, confusion matrices

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:04:11 UTC from IEEE Xplore. Restrictions apply.

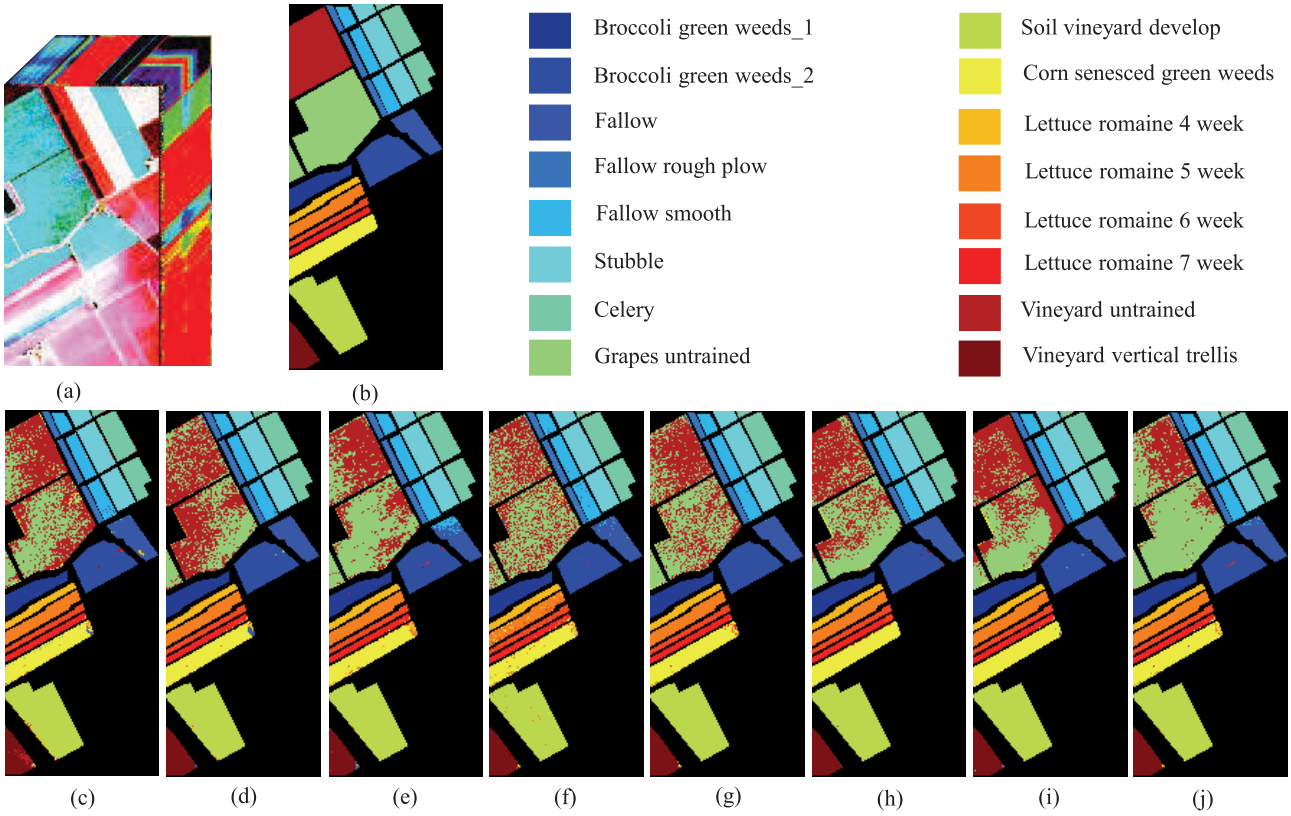


Fig. 3. Classification maps of different methods for Salinas: (a) Given HSI; (b) ground truth; (c) RAW; (d) EMAP; (e) IFRF; (f) LGIDL; (g) LRDLSR; (h) RPNet; (i) CNN-AL-MRF; and (j) Proposed DLRL.

Confusion Matrix		Predicted	
		Grapes untrained	Vineyard untrained
Real	Grapes untrained	3211	1074
	Vineyard untrained	2313	2505

(a) RAW

Confusion Matrix		Predicted	
		Grapes untrained	Vineyard untrained
Real	Grapes untrained	2695	1088
	Vineyard untrained	2875	2491

(b) EMAP

Confusion Matrix		Predicted	
		Grapes untrained	Vineyard untrained
Real	Grapes untrained	4194	1082
	Vineyard untrained	1368	2497

(c) IFRF

Confusion Matrix		Predicted	
		Grapes untrained	Vineyard untrained
Real	Grapes untrained	3451	1487
	Vineyard untrained	2119	2092

(d) LGIDL

Confusion Matrix		Predicted	
		Grapes untrained	Vineyard untrained
Real	Grapes untrained	3295	1179
	Vineyard untrained	2311	2400

(e) LRDLSR

Confusion Matrix		Predicted	
		Grapes untrained	Vineyard untrained
Real	Grapes untrained	3577	899
	Vineyard untrained	1993	2680

(f) RPNet

Confusion Matrix		Predicted	
		Grapes untrained	Vineyard untrained
Real	Grapes untrained	3048	531
	Vineyard untrained	2522	3031

(g) CNN-AL-MRF

Confusion Matrix		Predicted	
		Grapes untrained	Vineyard untrained
Real	Grapes untrained	4992	1553
	Vineyard untrained	570	2026

(h) DLRL

Fig. 4. Confusion matrices of different methods to list the number of pixels that are correctly and incorrectly classified in "grapes untrained" and "vineyard untrained" classes.(a) RAW, (b) EMAP, (c) IFRF, (d) LGIDL, (e) LRDLSR, (f) RPNet, (g) CNN-AL-MRF, and (h) DLRL.

for the classification maps in Fig. 3 on these two classes are calculated, as shown in Fig. 4, where list correct classification numbers and error numbers of these two classes. As can be seen from Fig. 4, DLRL (7018) achieves more total correct number on the "Grapes untrained" and "Vineyard untrained" classes compared with numbers for RAW (5716), EMAP (5186), IFRF (6691), LGIDL (5543), LRDLSR (5695), RPNet (6257), and CNN-AL-MRF (6079). What's more, in Fig. S1, map results of RAW, EMAP, IFRF, LGIDL, LRDLSR, RPNet, and CNN-AL-MRF include serious classification errors for "Bitumen" on the white dashed region in Fig. S1(c)–(i). Whereas such errors are avoided in Fig. S1(j) obtained by DLRL. In Fig. S2, we can obviously observe that white dashed regions of those compared methods [see Fig. S2(c)–(i)]

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:04:11 UTC from IEEE Xplore. Restrictions apply.

TABLE I

METRIC RESULTS OF DIFFERENT METHODS FOR SALINAS. THE BEST OA, AA, AND KAPPA VALUES ARE MARKED IN **BOLD**

Class number/name	Training/ Testing	Classification Methods							
		RAW	EMAP	IFRF	LGIDL	LRDLSR	RPNet	CNN-AL-MRF	DLRL
1 Broccoli green weeds_1	25 /1034	99.01	99.57	99.80	99.37	99.16	99.37	98.54	100
2 Broccoli green weeds_2	25/1838	96.79	99.79	99.12	99.34	100	99.23	99.12	99.12
3 Fallow	25 /973	98.73	98.80	81.17	95.94	99.57	99.79	99.79	96.37
4 Fallow rough plow	25 /682	98.12	98.62	100	98.10	100	100	100	100
5 Fallow smooth	25 /1319	98.76	99.80	98.76	97.96	99.53	99.06	97.80	99.53
6 Stubble	25 /1944	99.20	99.66	99.69	99.48	99.79	100	99.59	99.69
7 Celery	25 /1764	97.77	98.31	99.77	99.43	99.53	98.97	98.39	99.66
8 Grapes untrained	25 /5570	57.65	48.38	75.27	61.96	59.16	64.22	54.72	89.62
9 Soil vineyard develop	25 /3051	97.45	99.42	99.55	98.95	99.93	99.80	98.19	99.80
10 Corn senesced green weeds	25 /1614	90.51	93.56	90.63	89.75	93.67	97.22	99.37	94.56
11 Lettuce romaine 4 week	25 /524	91.32	92.66	99.21	97.52	98.35	99.59	99.12	99.59
12 Lettuce romaine 5 week	25 /948	100	100	99.78	85.49	100	100	100	100
13 Lettuce romaine 6 week	25 /450	99.51	99.09	98.12	83.25	99.01	99.47	98.03	99.51
14 Lettuce romaine 7 week	25 /527	96.71	91.54	95.26	83.54	97.53	91.36	85.00	94.24
15 Vineyard untrained	25 /3579	69.98	69.60	69.76	58.46	67.05	74.87	84.71	56.60
16 Vineyard vertical trellis	25 /888	99.12	99.31	96.48	98.56	98.56	99.76	99.04	99.28
OA(%)		84.34	83.51	88.98	83.93	88.03	89.69	89.60	**91.14**
AA(%)		93.16	87.22	93.90	90.44	94.42	94.98	94.46	**95.47**
Kappa×100		82.47	81.61	87.73	82.10	86.68	88.52	88.45	**90.08**

have much gap with ground truth [see Fig. S2(b)]. However, the classification result on white dashed region of DLRL in Fig. S2(j) achieves great consistency with that of ground truth.

More specifically, we present metric results of different classification methods for Salinas, Pavia Center, and Botswana in Table I and Tables S1 and S2, in which the best values of OA, AA, and Kappa are marked in **bold**. From these three tables, it can be seen that our proposed DLRL method achieves the best values among all OA, AA, and Kappa results.

Based on the above experimental results and related discussion, we conclude that the proposed DLRL method obviously outperforms RAW, EMAP, IFRF, LGIDL, LRDLSR, RPNet, and CNN-AL-MRF. The reason for this is that, for RAW, EMAP, IFRF, RPNet, and CNN-AL-MRF, their features used for classifiers are extracted with disorder distribution. A feature vector may be very close to the ones from other classes, which will reduce the accuracy of classification. Although LGIDL and LRDLSR with LRR have clustering property, clustered features are yielded without the ideal classwise block-diagonal structure, which induce error-rate increase of classification. However, for the proposed DLRL method, the classwise block-diagonal structure is used to obtain a diagonalized-feature representation form of samples, which has order feature distribution with strong discriminant ability. These features are easily classified to calculate good classification maps. Furthermore, a tradeoff of collaboration with Frobenius norm and sparsity with l_1 norm regularizations is sought to obtain both correlations among HSI pixels and characteristics of each pixel, which can fully use HSI information for high-quality classification.

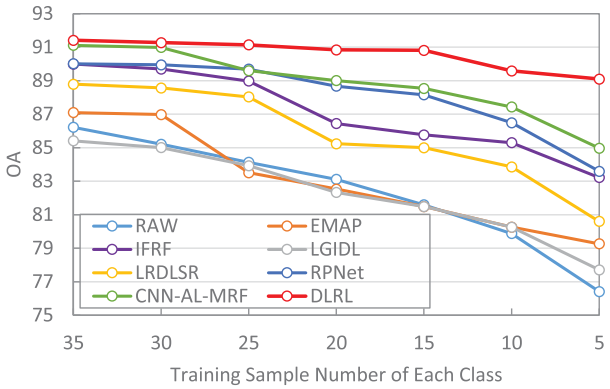


Fig. 5. Stability comparison of different classification methods on Salinas with different training sample number.

E. Discussion

In this section, we will discuss the classification stability for training sample number, effectiveness of the classwise regularization, effectiveness of sparsity, and effectiveness of collaboration in model (3). It should be noted that in the following experimental discussions, metrics OA and Kappa are employed to measure results, and the Salinas dataset is used for testing. All listed results are obtained by averaging over five experiments.

1) Stability of DLRL for Training Sample Number: To explore the stability of the proposed DLRL method, we make comparison with RAW, EMAP, IFRF, LGIDL, LRDLSR, RPNet, and CNN-AL-MRF on different training sample numbers, which are selected from the range of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:04:11 UTC from IEEE Xplore. Restrictions apply.

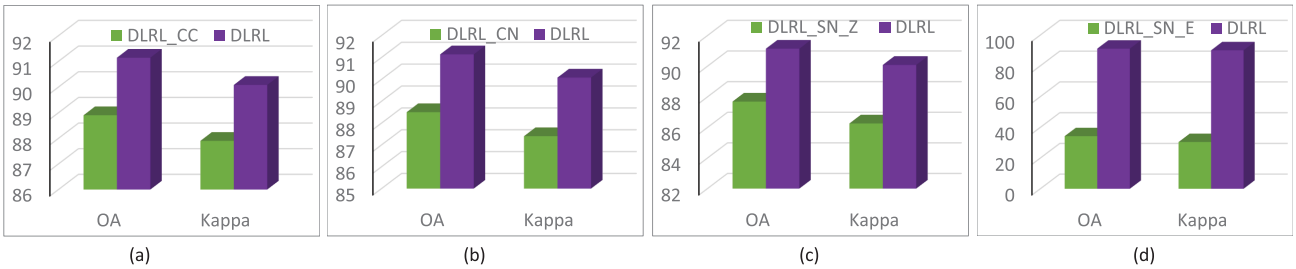


Fig. 6. DLRL is compared with (a) DLRL without classwise regularization $\sum_{i=1}^{C} \|_a Z_{\bullet i}\|_F^2$; (b) DLRL without collaboration regularization $\|Z\|_F^2$; (c) DLRL without sparsity regularization $\|Z\|_1$; (d) DLRL without sparsity regularization $\|E\|_1$.

[5, 10, 15, 20, 25, 30, 35]. The stability comparison of different methods are visually shown in Fig. 5. As can be found in this figure, the OA and Kappa results of all classification methods keep decreasing with training sample number increasing. It should be noted that in such number range (*i.e.*,$5-35$), the result change rate of DLRL is much lower than that of other seven methods. This indicates that the proposed DLRL method includes a better stability compared with RAW, EMAP, IFRF, LGIDL, LRDLSR, RPNet, and CNN-AL-MRF.

2) Effectiveness of Classwise Regularization: In the constructed model (3), classwise regularization $\sum_{i=1}^{C} \|_a Z_{\bullet i}\|_F^2$ is added to capture the classwise block-diagonal structure of HSIs. To validate the effectiveness of such classwise regularization, we compare DLRL with the DLRL form without classwise regularization (denoted by DLRL_CC), with results shown in Fig. 6(a). As can be seen in Fig. 6(a), the OA and Kappa values of DLRL_CC are obviously smaller than those of DLRL. This implies that classwise regularization promotes the classification accuracy of DLRL. In other words, the classwise regularization is effective in DLRL to assist in improving classification performance.

3) Effectiveness of Collaboration With Frobenius Norm Regularization: In model (3), we exploit the Frobenius norm regularization to investigate the correlations among samples, which is called collaboration for increased classification performance. For demonstrating its effectiveness, the proposed DLRL method is compared with the DLRL form without collaboration regularization (denoted by DLRL_CN). The comparison results are visually presented in Fig. 6(b). From this figure, we can observe that DLRL achieves larger OA and Kappa values compared with DLRL_CN. Therefore, it can be concluded that collaboration with Frobenius norm regularization included in DLRL is effective to describe correlations among samples for the performance improvement.

4) Effectiveness of Sparsity With l_1 Norm Regularization: There exist two sparse regularization terms with l_1 norm in model (3), including $\|Z\|_1$ and $\|E\|_1$. The $\|Z\|_1$ is used to impose the sparsity of LRR, and the $\|E\|_1$ is employed for noise suppression. To explore effectiveness of $\|Z\|_1$ and $\|E\|_1$, we compare DLRL with the following two forms: 1) DLRL without $\|Z\|_1$ (denoted by DLRL_SN_Z) and 2) DLRL without $\|E\|_1$ (denoted by DLRL_SN_E). On the one hand, as can be found in Fig. 6(c), DLRL_SN_Z obtains the smaller OA and Kappa values than DLRL, which means the accuracy improvement achieved by the sparsity of LRR. On the other hand, from Fig. 6(d), the OA and Kappa results of DLRL_SN_E are worse than those of DLRL. This indicates the good noise suppression ability obtained by imposing sparsity with l_1 norm regularization. Suggested by the above-mentioned experimental results and related discussion, we validate the effectiveness of sparsity with l_1 norm regularization (*i.e.*, $\|Z\|_1$ and $\|E\|_1$).

V. Conclusion

In this work, we develop a DLRL model for HSI classification. To obtain more discriminative feature representation, we introduce a classwise regularization to capture the classwise block-diagonal structure of LRR, which diagonalizes the feature representation. In such structure representation, HSI pixels from the same class are clustered into the same subspace, which assists to easily and correctly classify HSIs. In addition, we seek a tradeoff of sparsity and collaboration to simultaneously describe information of each spectral pixel and correlations among different pixels. More information can be thus used for classification. We evaluate the proposed DLRL method on three hyperspectral datasets including Salines, Pavia Center, and Botswana, with experimental results validating its effectiveness in HSI classification.

References

[1] C. Deng, X. Liu, C. Li, and D. Tao, "Active multi-kernel domain adaptation for hyperspectral image classification," *Pattern Recognit.*, vol. 77, pp. 306–315, 2018.

[2] C. Xing, M. Wang, C. Dong, C. Duan, and Z. Wang, "Joint sparse-collaborative representation to fuse hyperspectral and multispectral images," *Signal Process.*, vol. 173, Aug. 2020, Art. no. 107585.

[3] L. Zhang, L. Zhang, and B. Du, "Deep learning for remote sensing data: A technical tutorial on the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 22–40, Jun. 2016.

[4] K. Fukunaga and J. M. Mantock, "Nonparametric discriminant analysis," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. PAMI-5, no. 6, pp. 671–678, Nov. 1983.

[5] J. B. Lee, A. S. Woodyatt, and M. Berman, "Enhancement of high spectral resolution remote-sensing data by a noise-adjusted principal components transform," *IEEE Trans. Geosci. Remote Sens.*, vol. 28, no. 3, pp. 295–304, May 1990.

[6] W. Li, S. Prasad, J. E. Fowler, and L. M. Bruce, "Locality-preserving dimensionality reduction and classification for hyperspectral image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 4, pp. 1185–1198, Apr. 2012.

[7] L. He, J. Li, C. Liu, and S. Li, "Recent advances on spectral–spatial hyperspectral image classification: An overview and new guidelines," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1579–1597, Mar. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:04:11 UTC from IEEE Xplore. Restrictions apply.

[8] F. Tsai and J.-S. Lai, "Feature extraction of hyperspectral image cubes using three-dimensional gray-level cooccurrence," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 6, pp. 3504–3513, Jun. 2013.

[9] J. A. Benediktsson, M. Pesaresi, and K. Arnason, "Classification and feature extraction for remote sensing images from urban areas based on morphological transformations," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 9, pp. 1940–1949, Sep. 2003.

[10] P. R. Marpu, M. Pedergnana, M. D. Mura, J. A. Benediktsson, and L. Bruzzone, "Automatic generation of standard deviation attribute profiles for spectral–spatial classification of remote sensing data," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 2, pp. 293–297, Mar. 2013.

[11] W. Liao, R. Bellens, A. Pizurica, W. Philips, and Y. Pi, "Classification of hyperspectral data over urban areas using directional morphological profiles and semi-supervised feature extraction," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 4, pp. 1177–1190, Aug. 2012.

[12] Z. Y. Lv, P. Zhang, J. A. Benediktsson, and W. Z. Shi, "Morphological profiles based on differently shaped structuring elements for classification of images with very high spatial resolution," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 12, pp. 4644–4652, Dec. 2014.

[13] Z. Ye, S. Prasad, W. Li, J. E. Fowler, and M. He, "Classification based on 3-D DWT and decision fusion for hyperspectral image analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 173–177, Jan. 2014.

[14] B. Demir and S. Erturk, "Empirical mode decomposition of hyperspectral images for support vector machine classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4071–4084, Nov. 2010.

[15] L. Zhang, L. Zhang, D. Tao, and X. Huang, "Tensor discriminative locality alignment for hyperspectral image spectral–spatial feature extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 1, pp. 242–256, Jan. 2013.

[16] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[17] J. M. Haut, M. E. Paoletti, J. Plaza, J. Li, and A. Plaza, "Active learning with convolutional neural networks for hyperspectral image classification using a new Bayesian approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6440–6461, Nov. 2018.

[18] Y. Xu, B. Du, F. Zhang, and L. Zhang, "Hyperspectral image classification via a random patches network," *ISPRS J. Photogramm. Remote Sens.*, vol. 142, pp. 344–357, Aug. 2018.

[19] X. Cao, J. Yao, Z. Xu, and D. Meng, "Hyperspectral image classification with convolutional neural network and active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4604–4616, Jul. 2020.

[20] M. Zhang, W. Li, and Q. Du, "Diverse region-based CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2623–2634, Jun. 2018.

[21] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Jan. 2020.

[22] O. Okwuashi and C. E. Ndehedehe, "Deep support vector machine for hyperspectral image classification," *Pattern Recognit.*, vol. 103, Jul. 2020, Art. no. 107298.

[23] B. Xi *et al.*, "Deep prototypical networks with hybrid residual attention for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3683–3700, 2020.

[24] A. Sellami, A. B. Abbes, V. Barra, and I. R. Farah, "Fused 3-D spectral-spatial deep neural networks and spectral clustering for hyperspectral image classification," *Pattern Recognit. Lett.*, vol. 138, pp. 594–600, Oct. 2020.

[25] S. Jia, H. Xie, and X. Deng, "Extended morphological profile-based Gabor wavelets for hyperspectral image classification," in *Proc. 24th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2018, pp. 782–787.

[26] C. Zheng and N. Wang, "Collaborative representation with k-nearest classes for classification," *Pattern Recognit. Lett.*, vol. 117, pp. 30–36, 2018.

[27] H. Zhang, J. Li, Y. Huang, and L. Zhang, "A nonlocal weighted joint sparse representation classification method for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2056–2065, Jun. 2014.

[28] X. Zhang, P. Weng, J. Feng, E. Zhang, and B. Hou, "Spatial-spectral classification based on group sparse coding for hyperspectral image," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2013, pp. 1745–1748.

[29] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Hyperspectral image classification using dictionary-based sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3973–3985, Oct. 2011.

[30] L. Fang, C. Wang, S. Li, and J. A. Benediktsson, "Hyperspectral image classification via multiple-feature-based adaptive sparse representation," *IEEE Trans. Instrum. Meas.*, vol. 66, no. 7, pp. 1646–1657, Jul. 2017.

[31] Z. He, L. Liu, R. Deng, and Y. Shen, "Low-rank group inspired dictionary learning for hyperspectral image classification," *Signal Process.*, vol. 120, pp. 209–221, Mar. 2016.

[32] Z. He, L. Liu, S. Zhou, and Y. Shen, "Learning group-based sparse and low-rank representation for hyperspectral image classification," *Pattern Recognit.*, vol. 60, pp. 1041–1056, Dec. 2016.

[33] Z. Chen, X.-J. Wu, and J. Kittler, "Low-rank discriminative least squares regression for image classification," *Signal Process.*, vol. 173, Aug. 2020, Art. no. 107485.

[34] P. Ji, T. Zhang, H. Li, M. Salzmann, and I. Reid, "Deep subspace clustering networks," in *Proc. Conf. Workshop Neural Inf. Process. Syst.*, 2017, pp. 24–33.

[35] G. Liu, Z. Lin, and Y. Yu, "Robust subspace segmentation by low-rank representation," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2010, pp. 663–670.

[36] Q. Qu, N. M. Nasrabadi, and T. D. Tran, "Abundance estimation for bilinear mixture models via joint sparse and low-rank representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 4404–4423, Jul. 2014.

[37] Y. Yuan, M. Fu, and X. Lu, "Low-rank representation for 3D hyperspectral images analysis from map perspective," *Signal Process.*, vol. 112, pp. 27–33, Jul. 2015.

[38] Y.-Q. Zhao and J. Yang, "Hyperspectral image denoising via sparse representation and low-rank constraint," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 1, pp. 296–308, Jan. 2015.

[39] C. Xing, Z. Wang, Q. Ouyang, C. Dong, and C. Duan, "Image fusion method based on spatially masked convolutional sparse representation," *Image Vis. Comput.*, vol. 90, Oct. 2019, Art. no. 103806.

[40] L. Zhang, M. Yang, and X. Feng, "Sparse representation or collaborative representation: Which helps face recognition?" in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 471–478.

[41] J.-F. Cai, E. J. Candès, and Z. Shen, "A singular value thresholding algorithm for matrix completion," *SIAM J. Optim.*, vol. 20, no. 4, pp. 1956–1982, Jan. 2010.

[42] J. Cai, Y. Lei, M. Chen, N. Wang, and S. Zhang, "Efficient solution of the SVD recommendation model with implicit feedback," *Scientia Sinica Informationis*, vol. 50, no. 10, pp. 1544–1558, 2020.

[43] Z. Akata, F. Perronnin, Z. Harchaoui, and C. Schmid, "Good practice in large-scale learning for image classification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 3, pp. 507–520, Mar. 2014.

[44] Y. Wu *et al.*, "Approximate computing of remotely sensed data: SVM hyperspectral image classification as a case study," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 12, pp. 5806–5818, Dec. 2016.

[45] J. A. Benediktsson, J. A. Palmason, and J. R. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005.

[46] X. Kang, S. Li, and J. A. Benediktsson, "Feature extraction of hyperspectral images with image fusion and recursive filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 6, pp. 3742–3752, Jun. 2014.



Changda Xing received the M.Sc. degree from the Anhui University of Science and Technology, Huainan, China, in 2016, and the Ph.D. degree from the Nanjing University of Aeronautics and Astronautics (NUAA), Nanjing, China, in 2020.

He is a Postdoctoral Fellow at NUAA. His main research interests are machine learning, pattern recognition, image fusion, and hyperspectral image analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:04:11 UTC from IEEE Xplore. Restrictions apply.



Meiling Wang received the M.Sc. degree from the Nanjing University of Information Science and Technology, Nanjing, China, in 2016, and the Ph.D. degree from the Nanjing University of Aeronautics and Astronautics, Nanjing, in 2020.

She is a Postdoc Research Fellow with the Nanjing University of Aeronautics and Astronautics. Her main research interests are pattern recognition, machine/deep learning, and brain imaging genetics.



Zhisheng Wang received the B.S. and Ph.D. degrees from Northwestern Polytechnical University, Xi'an, China, in 1999 and 2002, respectively.

He was a Postdoctoral Fellow with the Nanjing University of Aeronautics and Astronautics from 2002 to 2004. From 2006 to 2008, he worked as a Postdoctoral Fellow again with the National University of Defense Technology. In 2010, he was a Senior Visiting Scholar with Stockholm University, Stockholm, Sweden. He is a Professor working with the Nanjing University of Aeronautics and Astronautics. He has authored or coauthored over 100 refereed papers. His main research interests are pattern recognition, information fusion, and computer/machine vision.



Chaowei Duan received the M.Sc. degree from the Wuhan University of Technology, Wuhan, China, in 2008. He is pursuing the Ph.D. degree with the Nanjing University of Aeronautics and Astronautics, Nanjing.

His research interests include pattern recognition and image processing.



Yiliu Liu received the bachelor's degree from the China University of Mining and Technology, Xuzhou, China, in 2019. He is pursuing the master's degree with the Nanjing University of Aeronautics and Astronautics, Nanjing, China.

His main interests include objection detection and recognition, image processing, and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:04:11 UTC from IEEE Xplore. Restrictions apply.